

5 Probability and Random Variables

This is an introduction, we'll take this to hidden Markov models etc.

Any recording of data is not going to be deterministic, there will be noise and uncertainty etc.

We have a sample space, S : the set of all possible outcomes.

Inside there are sample points, which are single outcomes. There is a set of possible unique outcomes (even numbers of a dice roll, under three, or whatever) that we call an "event".

Conditional probability:

$P(A|B)$ is the probability of getting A given B .

$$\begin{aligned}P(A|B) &= \frac{P(A \cap B)}{P(B)} \\P(A \cap B) &= P(A)P(B|A) \\P(A|B) &= \frac{P(A)P(B|A)}{P(B)} \quad (\text{bayes rule})\end{aligned}$$

Independence:

$$\begin{aligned}P(A|B) &= P(A) \\P(A \cap B) &= P(A)P(B)\end{aligned}$$

Lets consider rolling two dice.

$$\begin{aligned}A &= \text{sum eeb} \\B &= \text{sum} \\&= 1 \text{ roll is } 1\end{aligned}$$

Are A and B independent? A and ?

Roll 1, Roll 2

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$\begin{aligned}P(A|B) &= \frac{P(A, B)}{P(B)} = \frac{3}{6} \\P(A) &= \frac{P(A,)}{P()} = \frac{1}{2}\end{aligned}$$

Random Variables

We assign a number to an outcome as a number, and ask the probability that the number is equal to some value. We have probability mass distributions, and cumulative probability distributions.